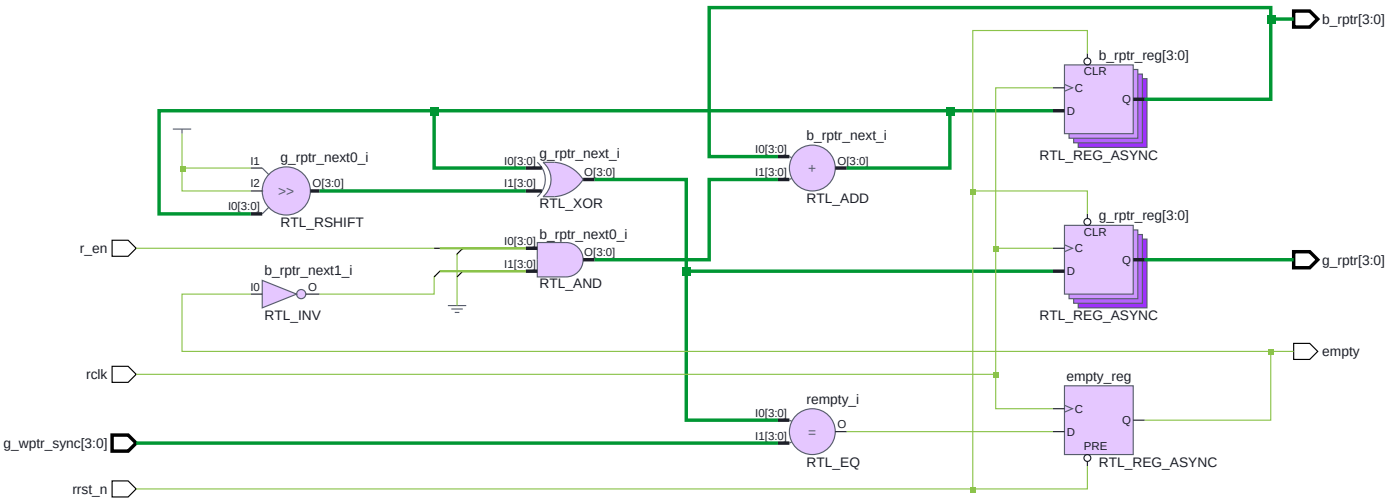
b_rptr[3:0]
b_rptr_reg[3:0]
CLR
C
D
Q
RTL_REG_ASYNC
g_rptr_next0_i
I1
I2
I0[3:0]
>>
O[3:0]
RTL_RSHIFT
g_rptr_next_i
I0[3:0]
I1[3:0]
O[3:0]
RTL_XOR
b_rptr_next_i
+
RTL_ADD
r_en
b_rptr_next0_i
RTL_AND
b_rptr_next1_i
I0
O
RTL_INV
g_rptr_reg[3:0]
g_rptr[3:0]
empty
rclk
empty_reg
PRE
rempty_i
=
RTL_EQ
g_wptr_sync[3:0]
rrst_n